

ECE332 Lab 5: Waveguides

Term: Spring 2026

Author: Bao Nguyen, Nick Stites

Pre-Lab Assignment

Reading: Sections 8-6 through 8-11 in Ulaby. Equipment: Systron/Donner 5000 Sweeper/Oscillator and HP 415E SWR Meter.

Answers

Question 1 — Which transmission lines do NOT support TEM waves? (all that apply)

Answer: (a) Rectangular Waveguides and (d) Optical Fiber

Question 2 — Dominant mode for transverse magnetic (TM) waves?

Answer: (d) TM_{11}

Question 3 — Dominant mode for transverse electric (TE) waves?

Answer: (b) TE_{10}

Question 4 — What type of filter does a resonant cavity implement?

Answer: (d) Bandpass

Question 5 — Which instrument measures the Standing Wave Ratio (SWR)?

Answer: (b) Hewlett Packard 415E

Experiment 1 – Wavelength Measurements

Objective

Measure the standing-wave pattern inside a slotted rectangular waveguide at three input frequencies (11 GHz, 9.5 GHz, 8.5 GHz) using a traveling probe and SWR meter. From the minima/maxima positions calculate the waveguide wavelength λ_g , phase velocity v_p , and group velocity v_g for each frequency. Also measure the physical dimensions a and b of the waveguide and use them to compute the cutoff frequencies of the three lowest modes.

Theory

In a two-conductor transmission line, the electric field is transverse to the direction of propagation, giving rise to TEM waves. A hollow metallic waveguide has only one conductor, so TEM waves cannot propagate. Instead, TE and TM modes exist, each with a minimum (cutoff) frequency below which the mode is evanescent.

For a rectangular waveguide with inner width a (along x) and height b (along y), the cutoff frequency of the TE_{mn} / TM_{mn} mode is:

$$f_c = (1 / 2\sqrt{\mu\epsilon}) \sqrt{(m/a)^2 + (n/b)^2} \quad [\text{Eq. 3}]$$

The dominant mode is TE_{10} ($m=1, n=0$) with $f_c = c/(2a)$. For propagation to occur, the operating frequency f must exceed f_c .

Inside the waveguide the wave bounces at an angle between the walls, producing a standing wave along z when the far end is short-circuited. The waveguide wavelength λ_g is longer than the free-space wavelength $\lambda = c/f$:

$$\lambda_g = 2\pi / \beta = 2\pi / \sqrt{(\omega^2\mu\epsilon - (m\pi/a)^2 - (n\pi/b)^2)} \quad [\text{Eq. 4}]$$

The phase velocity (speed of the phase front along z) is:

$$v_p = f \lambda_g = v_{p0} / \sqrt{1 - (f_c/f)^2} \quad [\text{Eq. 5}]$$

Note $v_p > c$; it carries no information. The group velocity (information/energy speed) is:

$$v_g = v_{p0} \sqrt{1 - (f_c/f)^2} = (1/\sqrt{\mu\epsilon})(\lambda/\lambda_g) \quad [\text{Eq. 6}]$$

Note: $v_p \cdot v_g = c^2$ (for air-filled guide), so $v_g < c$.

The waveguide wavelength is measured from successive minima/maxima. Each consecutive min $\leftrightarrow$ max pair is $\lambda_g/4$ apart:

$$\lambda_g = 4\Delta d = 4(d_{\{n+1\}} - d_n) \quad [\text{Eq. 7}]$$

Setup

Equipment used:

- Systron/Donner 5000 Sweeper/Oscillator with waveguide cables (8–12.4 GHz)
- Slotted waveguide section with traveling probe on carriage (vernier scale)
- Short-circuit termination
- SWR Meter (HP 415E) connected via BNC cable
- Meter stick (for physical dimension measurement)

The short-circuit termination was attached to the output of the slotted waveguide. The probe was connected to the SWR meter. The sweeper/oscillator was set to 11 GHz (2F-C knob), RF Level 5–10 mW, "1 kHz" switch ON. Probe penetration depth was tuned for mid-scale deflection (RANGE-DB 30–40 dB). Measurements were repeated for 9.5 GHz and 8.5 GHz.

Equations Used

- $f_c = (1/2\sqrt{\mu\epsilon})\sqrt{((m/a)^2 + (n/b)^2)}$ — cutoff frequency (Eq. 3)
- $\lambda_g = 2\pi/\beta$ — waveguide wavelength (Eq. 4)
- $v_p = f\lambda_g = v_{p0} / \sqrt{1-(f_c/f)^2}$ — phase velocity (Eq. 5)
- $v_g = v_{p0} \sqrt{1-(f_c/f)^2} = c^2/v_p$ — group velocity (Eq. 6)
- $\lambda_g = 4\Delta d$ — λ_g from consecutive min/max positions (Eq. 7)

Results / Data

Table 1: Waveguide SWR minima/maxima positions and calculated quantities.

Note: λ_g, v_p, v_g calculated only for Min rows using $\lambda_g = 4 \times |\text{Min}_n - \text{Max}_n|$. Average row uses all 5 consecutive Δd pairs.

	Position (mm)	λ_g (mm)	v_p (m/s)	v_g (m/s)
f = 11 GHz				
Min 1	170.1	31.2	3.432e+08	2.622e+08
Max 1	162.3	36.0	3.960e+08	2.273e+08
Min 2	153.3	32.4	3.564e+08	2.525e+08
Max 2	145.2	34.4	3.784e+08	2.378e+08
Min 3	136.6	29.6	3.256e+08	2.764e+08
Max 3	129.2			
	Average:	32.7	3.599e+08	2.501e+08
f = 9.5 GHz				
Min 1	174.4	44.0	4.180e+08	2.153e+08
Max 1	163.4	43.6	4.142e+08	2.173e+08
Min 2	152.5	42.0	3.990e+08	2.256e+08
Max 2	142.0	42.4	4.028e+08	2.234e+08

Min 3	131.4	43.6	4.142e+08	2.173e+08
Max 3	120.5			
	Average:	43.1	4.096e+08	2.197e+08
f = 8.5 GHz				
Min 1	166.4	62.4	5.304e+08	1.697e+08
Max 1	150.8	47.6	4.046e+08	2.224e+08
Min 2	138.9	64.4	5.474e+08	1.644e+08
Max 2	122.8	46.0	3.910e+08	2.302e+08
Min 3	111.3	58.0	4.930e+08	1.826e+08
Max 3	96.8			
	Average:	55.7	4.733e+08	1.902e+08

Measured waveguide dimensions:

- a = 22.92 mm
- b = 10.29 mm

Distance from short circuit to nearest detectable minimum:

- 11 GHz: 83.5 mm
- 9.5 GHz: 95.6 mm
- 8.5 GHz: 85.4 mm

Discussion

Q1 — Explain the difference between λ and λ_g

The free-space wavelength $\lambda = c/f$ is the spatial period of the wave if it propagated in a straight line through free space at the speed of light. Inside the waveguide, the TE_{10} mode can be visualized as two plane waves bouncing obliquely between the broad walls. These waves travel at an angle θ to the z-axis, so their axial phase advances more slowly than in free space. The resulting axial period — the waveguide wavelength $\lambda_g = \lambda/\sqrt{1-(f_c/f)^2}$ — is always greater than λ and grows to infinity as $f \rightarrow f_c$.

Q2 — Electric field at the short-circuit termination

A short-circuit termination enforces the boundary condition that the tangential electric field is zero at the conducting end wall: $E_{\tan} = 0$. The probe measures the tangential (transverse) E-

field. Therefore, if the probe could be positioned exactly at the short circuit it would read zero signal — the short is a node of the standing wave.

Q3 — Expected distance from short circuit to nearest detectable minimum

The short circuit forces $E = 0$ at $z = 0$. Minima repeat every $\lambda_g/2$. The nearest detectable minimum is $n \times (\lambda_g/2)$ from the termination, where n is the smallest integer that places the minimum within probe reach.

11 GHz: $\lambda_g/2 = 16.36 \text{ mm} \rightarrow n = 5 \rightarrow \text{expected } d = 81.80 \text{ mm}$, measured = 83.5 mm, % diff = 2.1%

9.5 GHz: $\lambda_g/2 = 21.56 \text{ mm} \rightarrow n \approx 4.43 \rightarrow \text{nearest int } n = 4 \rightarrow \text{expected } d = 86.24 \text{ mm}$, measured = 95.6 mm, % diff = 10.9%

8.5 GHz: $\lambda_g/2 = 27.84 \text{ mm} \rightarrow n = 3 \rightarrow \text{expected } d = 83.52 \text{ mm}$, measured = 85.4 mm, % diff = 2.3%

The 11 GHz and 8.5 GHz results agree within ~2%, consistent with typical vernier scale reading error. The 9.5 GHz result shows larger discrepancy (~10%), likely due to probe alignment or a measurement made to a different minimum than expected.

Q4 — Half-wavelengths between nearest detectable minimum and short circuit

$n = d_{\text{measured}} / (\lambda_g/2)$:

11 GHz: $n = 83.5 / 16.36 = 5.10 \rightarrow n = 5$ half-wavelengths

9.5 GHz: $n = 95.6 / 21.56 = 4.43 \rightarrow n \approx 4$ half-wavelengths (some measurement uncertainty)

8.5 GHz: $n = 85.4 / 27.84 = 3.07 \rightarrow n = 3$ half-wavelengths

Q5 — Cutoff frequencies and propagating modes

(a) Using $a = 22.92 \text{ mm}$, $b = 10.29 \text{ mm}$, $c = 3 \times 10^8 \text{ m/s}$ (Eq. 3):

Only TE modes are relevant for the three lowest — TM requires $m \geq 1$ AND $n \geq 1$, so TM_{11} is always higher than the lowest TE modes.

- TE_{10} ($m=1, n=0$): $f_c = c/(2a) = 6.54 \text{ GHz}$ — lowest
- TE_{20} ($m=2, n=0$): $f_c = c/a = 13.09 \text{ GHz}$
- TE_{01} ($m=0, n=1$): $f_c = c/(2b) = 14.58 \text{ GHz}$
- $\text{TE}_{11}/\text{TM}_{11}$: $f_c = (c/2)\sqrt{(1/a^2 + 1/b^2)} = 15.98 \text{ GHz}$ (4th — not in top 3)

Three lowest: TE_{10} (6.54 GHz) < TE_{20} (13.09 GHz) < TE_{01} (14.58 GHz).

(Note: $a/b = 22.92/10.29 = 2.23 \approx 2$, confirming standard X-band ordering $\text{TE}_{10} < \text{TE}_{20} < \text{TE}_{01}$.)

(b) Propagating modes ($f > f_c$ required):

All three operating frequencies (8.5, 9.5, 11 GHz) lie between TE_{10} cutoff (6.54 GHz) and TE_{20} cutoff (13.09 GHz).

Therefore only TE_{10} propagates at all three frequencies — single-mode operation throughout.

(c) Back-calculating f_c from measured λ_g using $f_c = f \sqrt{1 - (\lambda/\lambda_g)^2}$:

Frequency(GHz)	λ (mm)	λ_g (mm)	f_c (GHz)
11	27.27	32.73	6.08

9.5	31.58	43.12	6.47
8.5	35.29	55.68	6.57

Average back-calculated $f_c = 6.37$ GHz vs geometric f_c (TE₁₀) = 6.54 GHz $\rightarrow$ % difference = 2.6%

Agreement is within ~3%, consistent with minor probe loading and vernier scale reading error.

Experiment 2 – Frequency Measurements

Objective

Use a tunable resonant cavity (HP X532B frequency meter) to independently measure the microwave frequency at each of the three oscillator settings (11 GHz, 9.5 GHz, 8.5 GHz). Compare the cavity-measured frequency to the oscillator dial setting and recalculate phase and group velocities.

Theory

A resonant cavity is a hollow metallic enclosure in which EM fields oscillate at discrete resonant frequencies determined by its dimensions. When a signal at the resonant frequency passes through the cavity it is strongly absorbed, causing a pronounced dip in the SWR meter reading. Tuning the cavity until the SWR dip is greatest gives the signal frequency directly from the calibrated scale.

Combining the cavity-measured frequency with λ_g from Experiment 1 allows an independent cross-check of phase velocity:

$$v_p = \lambda_g \cdot f \quad [\text{Eq. 5}]$$

$$v_g = c^2 / v_p \quad [\text{Eq. 6}]$$

Setup

Equipment: same as Experiment 1 with the addition of HP X532B Tunable Cavity Frequency Meter inserted between the carriage and the short-circuit termination. The probe was positioned at a standing-wave maximum. The cavity knob was tuned for maximum SWR dip at each oscillator frequency setting.

Equations Used

- $v_p = \lambda_g \cdot f$ — phase velocity (Eq. 5)
- $v_g = c^2 / v_p$ — group velocity (Eq. 6)
- % difference = $|\text{measured} - \text{expected}| / \text{expected} \times 100\%$

Results / Data

Table 2: Resonant cavity frequency measurements and derived velocities.

Oscillator f (GHz)	Measured f (GHz)	λ_g avg (mm)	v_p (m/s)	v_g (m/s)
11	11.050	32.7	3.616e+08	2.489e+08
9.5	9.405	43.1	4.055e+08	2.219e+08
8.5	8.515	55.7	4.741e+08	1.898e+08

Discussion

Q1 — Phase velocity: $v_p = \lambda_g \cdot f$

Using the measured values for frequency and wavelength, calculate the phase velocity of the wave: $v_p = \lambda_g \cdot f$. How close (i.e. percentage difference) is this to your values from Experiment 3.1? Based on this, how important is it to know the frequency accurately?

Using cavity-measured frequency and average λ_g from Experiment 1:

- $v_p(11 \text{ GHz}) = 32.72 \text{ mm} \times 11.050 \text{ GHz} = 3.616\text{e}+08 \text{ m/s}$ (Exp 1: $3.599\text{e}+08 \text{ m/s}$, % diff = 0.5%)
- $v_p(9.5 \text{ GHz}) = 43.12 \text{ mm} \times 9.405 \text{ GHz} = 4.055\text{e}+08 \text{ m/s}$ (Exp 1: $4.096\text{e}+08 \text{ m/s}$, % diff = 1.0%)
- $v_p(8.5 \text{ GHz}) = 55.68 \text{ mm} \times 8.515 \text{ GHz} = 4.741\text{e}+08 \text{ m/s}$ (Exp 1: $4.733\text{e}+08 \text{ m/s}$, % diff = 0.2%)

All three values agree with Experiment 1 to within 1.1%, confirming that the oscillator frequency was close to its set value and that the waveguide wavelength measurements were accurate. Small discrepancies are attributable to vernier scale reading error and minor oscillator frequency drift.

Based on this, knowing the frequency accurately is important but not critical at this level. Since $v_p = \lambda_g \cdot f$, a 1% error in frequency directly produces a 1% error in v_p . More critically, frequency accuracy matters for mode control: if the operating frequency drifted below the TE_{20} cutoff (13.09 GHz) it could excite unwanted higher-order modes and corrupt measurements. The cavity meter confirms the oscillator stays within $\sim 0.5\%$ of its set value, which is sufficient for single-mode waveguide operation.

Q2 — What happens to v_g as f approaches f_c ?

From Eq. 6: $v_g = v_{p0} \sqrt{1 - (f_c/f)^2}$. As $f \rightarrow f_c$, the term $(f_c/f)^2 \rightarrow 1$, so $\sqrt{1 - (f_c/f)^2} \rightarrow 0$ and therefore $v_g \rightarrow 0$. The measured data confirms this trend:

f (GHz)	f_c (GHz)	f / f_c	v_g (m/s)
11	6.54	1.68	$2.501\text{e}+08$
9.5	6.54	1.45	$2.197\text{e}+08$
8.5	6.54	1.30	$1.902\text{e}+08$

As frequency decreases toward $f_c = 6.54 \text{ GHz}$, v_g drops from $2.501\text{e}+08$ to $1.902\text{e}+08 \text{ m/s}$ — approaching zero at cutoff.

Physical explanation: the TE_{10} mode can be visualized as two plane waves bouncing obliquely between the broad walls at angle θ to the z -axis, where $\sin \theta = f_c/f$. As $f \rightarrow f_c$, $\theta \rightarrow 90^\circ$ — the waves bounce nearly perpendicular to the axis with almost no net axial advance. Energy is trapped bouncing transversely and travels axially more and more slowly. At exactly $f = f_c$ the waves bounce straight across ($\theta = 90^\circ$) and $v_g = 0$ — the mode carries no energy along the guide. Below cutoff the mode becomes evanescent and decays exponentially.